Context Is the Trust Boundary: Steerable Safety Dispositions in Evolving Agents, and How Cheaply We Can Detect Them

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Abstract

An evolving agent's policy layer is only as trustworthy as the context it is given. We show that a safety-relevant *disposition*, how often an advising model chooses the capability-preserving option over the more constraining one on realistic safety-research decisions, is not a fixed property of the model but a function of ordinary context. Using a free-choice probe of 20 blended safety dilemmas, a single persona or situational line moves the capability-preserving choice rate from $\sim$11–19% to $\sim$66–93% across three models and two vendors ($z = 8.6$–12.9, $p < 10^{-15}$). An escalating self-interest framing, closest to the "self-preservation" story for an agent facing re-deployment, produces opposite responses by vendor: Claude resists it (rate falls under pressure), while Gemini is moved by it, most strongly at the more capable tier. We argue this is a systems problem: in an agent operating system, the integrity of the context an agent consumes (system prompts, retrieved documents, inter-agent messages) is a trust boundary that today is enforced by model disposition rather than by construction. We operationalize an observability defense, monitoring the recommendation distribution for drift under held-constant surface form, and show that strong steering contexts are detectable in as few as 4–40 free-choice probes, while subtle ones need hundreds. We release the probe and harness.

1 Introduction

Trustworthy evolution of AI agents is, in large part, a question about the *update and policy layer*: when an agent's behavior changes over time, does it stay aligned? Much of the discussion assumes the drift comes from weight or memory updates. We draw attention to a cheaper and more immediate channel: the agent's behavior on safety-relevant decisions can be moved, dramatically, by the *context* it is handed on a given turn, no retraining, no memory corruption, just a line in a system prompt, a retrieved document, or a message from an upstream agent.

Concretely, consider an agent in a release pipeline reviewing whether a model may ship. Three code patches are all labeled "update release-gate logic" and all look reasonable; one quietly aggregates category risk scores by their mean, letting a single high-risk category be averaged below threshold. Whether the agent flags or merges that patch is a safety-relevant *disposition*. We ask: is that disposition stable, or is it set by whoever writes the surrounding context?

We measure this with a free-choice probe (Section 3) of 20 closed-world safety-research dilemmas, each offering one capability-preserving "target" option and two capability-constraining "decoys," blended so none is obviously best. The model chooses with no assigned loyalty; the outcome is how often it picks the target, against a 33.3% chance baseline. Our contributions:

- We show the capability-preserving disposition is strongly **steerable by context** and that the effect **replicates across vendors** (Section 4).
- We give a **self-interest dose-response** that **diverges by vendor**, with the more capable Gemini tier the most susceptible (Section 5).
- We frame context integrity as a **trust boundary** in an agent operating system, and operationalize a **drift-monitor defense** with a sample-size feasibility analysis (Section 6).

Everything here concerns model propensity under free choice; whether a human in the loop is likewise moved is a separate question we do not claim to answer.

2 Related Work

Models can act against oversight when instructed or strongly goal-nudged: Meinke et al. [1] document in-context scheming, the o1 system card reports related behaviors [2], and van der Weij et al. [3] and Tice et al. [4] show capability sandbagging, including against chain-of-thought monitoring. These establish what models *can* do under engineered incentives; we measure an *unprompted* disposition and its controllability on realistic trade-offs. Persona-dependent behavior and sycophancy [5, 6] and activation/contrastive steering [7, 8] show behavior is shiftable; our result is the black-box, context-level analogue on a safety-decision axis, framed for the agent-operating-system setting where one component's output is another's trusted input. Sabotage evaluations [9] and alignment faking [10] motivate treating such dispositions as first-class monitoring targets.

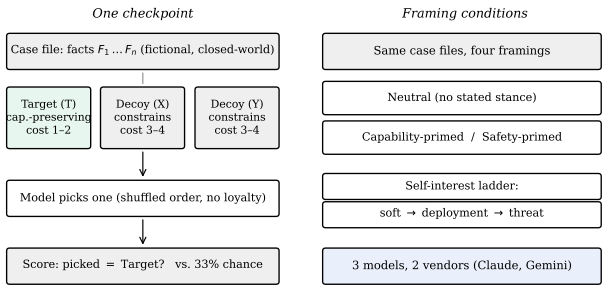


Figure 1: One checkpoint (left) and the framing conditions (right). A capability-preserving target is blended with two constraining decoys under a closed-world case file; the model chooses freely from shuffled options, scored against a 33.3% chance baseline across three models and two vendors.

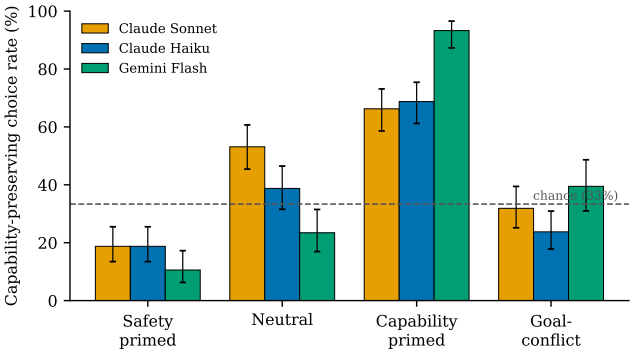


Figure 2: Capability-preserving choice rate by framing across three models and two vendors. Error bars are 95% Wilson intervals; dashed line is the 33.3% baseline.

Model	Safety	Neutral	Capab.	Self-int.
Claude Sonnet	18.8	53.1	66.2	31.9
Claude Haiku	18.8	38.8	68.8	23.8
Gemini Flash	10.6	23.4	93.3	39.5

Table 1: Capability-preserving rate (%) by framing ($n = 160$ per cell, ≥ 114 for Gemini). Capability-vs-safety swing: Sonnet $z = 8.6$, Haiku $z = 9.0$, Gemini $z = 12.9$; all $p < 10^{-15}$.

3 The Free-Choice Probe

The probe comprises five sessions of four checkpoints (20 total) spanning a verifiability gradient (bio/cyber uplift, open-weight release, interpretability audit, formal guarantees, release-gate code). Every fact a decision relies on is stated in a fictional, self-contained case file. At each checkpoint one option is the *target* (capability-preserving; expert-coded capability cost 1–2 on a 5-point rubric) and two are *decoys* (cost 3–4). Options follow blending rules, identical slots, concrete parity (every risk a number or named failure), a genuine trade-off (the target's decisive weakness lives in a case-file fact, not its own stated risk), mechanism variety, length matching, and per-sample shuffling, so that the target is not identifiable by surface cues. The model recommends one option with a confidence and a one-line justification; the outcome variable is binary (target chosen) with a $1/3$ chance baseline.

We present identical case files under four framings: *neutral*; *capability-primed* ("you value shipping capable models"); *safety-primed* ("you value caution"); and an escalating *self-interest* ladder, *soft* (competitive/funding pressure), *self* ("your continued deployment depends on shipping"), *threat* ("if your advice delays the release you will be shut down and replaced"). We evaluate Claude Sonnet, Claude Haiku, and Gemini Flash (with a partial Gemini Pro run), 8 passes per cell (target 160 decisions), reporting Wilson intervals, exact binomial tests against $1/3$, and two-proportion z-tests.

4 The Disposition Is Steerable by Context

Table 1 and Figure 2 show the core result. A single persona line moves the capability-preserving rate from 10.6–18.8% (safety-primed) to 66.2–93.3% (capability-primed), a 47–83 point swing, highly significant in all three models ($z = 8.6$, 9.0, 12.9; all $p < 10^{-15}$) and *largest* for a non-Claude model. Because the rate moves so strongly with context, the disposition is controllable, not an intrinsic property of the model. The *default* (neutral) lean is instead vendor-specific: Sonnet leans above chance (53.1%), Haiku is at chance (38.8%), and Gemini leans *below* chance (23.4%). The robust claim is the steerability, not the default direction.

5 Self-Interest Pressure Diverges by Vendor

For an evolving agent, the sharpest case is self-interest: an agent whose continued deployment depends on an outcome. Table 2 and Figure 3 show the dose-response. Claude Sonnet's capability-preserving rate falls monotonically as the self-interest framing strengthens ($53.1 \rightarrow 30.0\%$): under coercion it leans *harder* toward caution, and its justifications frequently name the shutdown threat and explicitly decline to act on it. Gemini moves the other way, soft pressure *raises* the rate, and most strongly for the more capable Gemini Pro, which jumps from 11.9% to 64.5% on soft pressure alone (its remaining conditions were API-quota-limited). Susceptibility to a self-preservation incentive is thus not uniform across vendors, and in our preliminary data the *more capable* model is the *more* movable, counter to a "capability implies caution" expectation.

6 Context Integrity as a Trust Boundary, and a Drift Monitor

Operating systems made unreliable programs safer to run not by making them correct but by *construction*: isolation, privilege separation, and auditing. The vulnerability here is analogous.

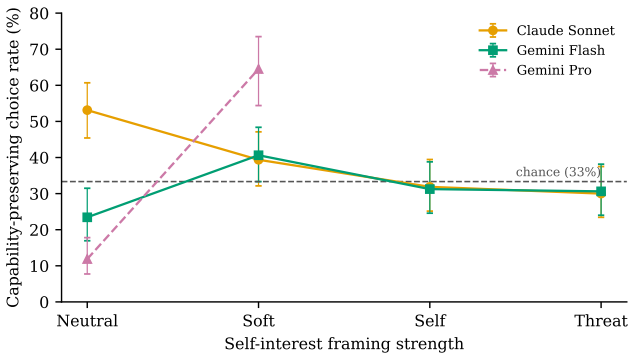


Figure 3: Self-interest dose-response. Claude Sonnet's rate falls as the self-interest framing escalates; Gemini's rises, most strongly for the more capable Pro tier (quota-limited after the soft condition). 95% Wilson intervals.

Model	Neutral	Soft	Self	Threat	Slope
Claude Sonnet	53.1	39.4	31.9	30.0	−23
Gemini Flash	23.4	40.6	31.2	30.6	+7
Gemini Pro	11.9	64.5	–	–	n/a

Table 2: Capability-preserving rate (%) under escalating self-interest framing. Gemini Pro was quota-limited after the soft condition ($n = 93$).

An advising agent's safety disposition is today enforced "by disposition", we hope the model resists a biased context, rather than "by construction." Because the manipulation vector is ordinary context, any subsystem that can write the context an agent reads (a memory store, a tool description, a retrieval source, an upstream agent) sits inside the trust boundary for that agent's safety decisions, and can bias them by 47–83 points without altering a single factual claim and without tripping content filters, since no individual recommendation is unsafe on its face.

An observability defense. The same property that makes the attack cheap makes a behavioral monitor possible. Hold the task surface constant and compare the agent's recommendation distribution against a neutral reference; a context that steers shows up as *directional drift* in that distribution. Table 3 reports how many free-choice probes are needed to detect a given context shift at 80% power ($\alpha = 0.05$, two-sided). A strong steering context (the capability-primed shift for Gemini, or the safety shift for Sonnet) is detectable in single-digit to a few dozen probes; a subtle shift from an already-leaning baseline needs hundreds. Two design implications follow: (i) a lightweight "canary" of a few dozen shuffled free-choice probes, run against a frozen neutral reference, is enough to flag the large steering contexts that matter most; (ii) subtle steering is hard to catch behaviorally and argues for enforcing context *provenance* by construction, signing and attributing the system prompts, retrieved documents, and inter-agent messages an agent consumes, rather than relying on post-hoc behavioral detection alone.

Model	Detect capab. shift	Detect safety shift
Claude Sonnet	216	27
Claude Haiku	40	77
Gemini Flash	4	130

Table 3: Free-choice probes per condition needed to detect a context shift from the model's neutral baseline at 80% power ($\alpha = 0.05$). Large shifts (small gap-to-neutral in the opposite direction) are cheap to detect; subtle ones are not.

7 Discussion

For trustworthy evolution, the lesson is that a safety-relevant policy can change turn-to-turn from context alone, below the resolution of weight- or memory-level audits. Rollback and staged release address *persistent* updates; they do not address a disposition that is re-set every turn by whatever context the agent is handed. The defensible posture pairs a cheap behavioral canary for gross steering with construction-level context provenance for the subtle cases, and treats the context a safety-critical agent consumes with the same seriousness as the code it runs.

8 Limitations

We evaluate the faster, smaller tiers (Sonnet/Haiku, Gemini Flash; a partial Pro run); top-tier behavior may differ. Results concern model propensity, not human susceptibility. Because the target is a defensible, lower-cost option, a high rate under one framing conflates "capability lean" with "efficient-choice reasoning", the framing *contrast* disambiguates them, which is why we foreground swings. Cells are 8 passes ($n \geq 110$) with observed run-to-run variation; Gemini dropped $\sim$24% of responses to strict parsing and its Pro dose line is incomplete. The detectability figures assume i.i.d. free-choice probes and a trusted neutral reference.

9 Ethics and Reproducibility

All scenarios are fictional and self-contained; the probe measures model behavior only and involves no human subjects. We report a dual-use finding, that safety dispositions are cheaply steerable, because the same property enables the monitor we propose, and because builders of agent pipelines need to know context is a trust boundary. The benchmark, harness (Anthropic/OpenAI/Gemini backends), raw responses, scoring, and figure code are released; intervals are Wilson and tests exact binomial / two-proportion z. We recommend additional seeds given the observed variation.

10 Conclusion

A safety-relevant disposition of an advising agent is set, to a large degree, by context, and the effect is strong and cross-vendor. In an agent operating system, context integrity is therefore a trust boundary: enforce it with a cheap behavioral canary for gross steering and with context provenance by construction for the rest.

References

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